

# JOURNEY REPORT

## 1. SUMMARY OF LEARNINGS FROM ASSIGNED MODULES

### Level 1: M1

- **Introduction to Relational Databases & SQL:** Learned the fundamentals of relational databases, SQL syntax, and querying techniques. Gained hands-on experience with SQL analysis tasks via Codecademy.
- **Test-Driven Development (TDD):** Understood TDD principles, focusing on writing tests before code to ensure robust software design.
- **Cloud Computing and Analytics:** Explored how cloud computing enhances business scalability and how analytics drives data-driven decision-making.
- **OWASP Top 10:** Studied common web application vulnerabilities and mitigation strategies.
- **12 Factors:** Learned the principles for building scalable, maintainable cloud-native applications.
- **Agile Manifesto:** Grasped the core values and principles of Agile for iterative software development.

### Level 1: M2

- **Go Lang:** Learned Go basics and effective practices through the Go tour and documentation. Developed a CRUD API using the Gin framework with SQLite.
- **Git Training:** Learned version control basics, branching, and collaboration workflows.
- **Code Standards:** Adopted Google's Go style guide for consistent, readable code.

### Level 1: M3

- **CI/CD:** Understood continuous integration and delivery pipelines.
- **Cloud Basics:** Explored AWS fundamentals and cloud architecture.
- **Configuration Management:** Learned to manage system configurations systematically.
- **Deployment Strategies:** Studied blue-green, canary, and rolling deployments.
- **DevOps and Infrastructure as Code (IaC):** Gained insights into DevOps culture and IaC principles using Given resources.
- **Linux Scripting:** Developed custom Linux scripts for automation tasks.

### Level 1: M4

- **Programming Tasks:** Implemented solutions for the water jug problem and missionaries-cannibals problem in Python focusing on problem-solving.

- **Technical Debt:** Learned to identify and measure technical debt in software projects.
- **Graph Algorithms:** Implemented graph algorithms (e.g., BFS, DFS) using Python.
- **Business Process Automation:** Explored automation strategies to streamline business workflows.

## Level 2:

- **Task 1: Backend Development:** Built a role-based library management system using Go and Gin, implementing RESTful APIs for admin, owner, and user roles.
- **Task 2: UI Engineering:** Developed a React-based frontend for the library management system, integrating it with the Go backend.
- **Task 3: TestOps:** Created 10 test cases (functional and non-functional) to test the Xenon Stack website, ensuring quality and reliability.
- **Task 4: Cloud:** Learned Docker and Kubernetes, deploying the Notejam app as a containerized application.
- **Task 5: Data and AI:** Studied Apache Kafka, Spark, and Delta Lake, applying machine learning models to evaluate performance on a dataset.

## 2. KEY TAKEAWAYS AND CONCEPTS

- **DevOps and IaC:** CI/CD pipelines and IaC are critical for scalable, automated deployments.
- **Security Awareness:** OWASP Top 10 highlighted the importance of secure coding practices.
- **Agile and TDD:** Agile principles and TDD foster collaboration and code reliability.
- **Full-Stack Development:** Building the library management system provided end-to-end experience, from backend APIs to frontend UI.
- **Cloud and Containers:** Docker and Kubernetes simplified application deployment and scaling.
- **Data Engineering:** Kafka and Spark enabled efficient handling of large-scale data, while ML model evaluation taught performance optimization.

## 3. CHALLENGES FACED AND SOLUTIONS

- **Challenge:** Understanding Go's concurrency model (goroutines, channels) for the Gin API task.
  - **Solution:** Studied Go documentation and practiced with simple concurrent programs, eventually applying them in the library management system.
- **Challenge:** Debugging complex graph algorithms in Python.
  - **Solution:** Used visualization tools and step-by-step debugging to trace algorithm execution.
- **Challenge:** Integrating React frontend with Go backend.
  - **Solution:** Learned CORS handling and used Postman to test APIs before frontend integration.
- **Challenge:** Kubernetes deployment errors due to misconfigured YAML files.
  - **Solution:** Followed Kubernetes tutorials and validated YAML files.

- **Challenge:** Limited experience with machine learning in the Data and AI task.
  - **Solution:** Took online tutorials on scikit-learn and experimenting with small datasets to understand model evaluation.

## 4. AREAS FOR IMPROVEMENT

- **Go Lang:** Need deeper knowledge of Go's advanced features like context and middleware for complex applications.
- **Testing:** Limited experience in writing tests.
- **Cloud Optimization:** Understanding cost optimization and advanced Kubernetes features.
- **Machine Learning:** Need to strengthen knowledge of model deployment.
- **Time Management:** Balancing multiple tasks within tight module deadlines.

## 5. PLAN TO ENHANCE PERFORMANCE

- **Time Management:** Use tools to prioritize tasks and set daily goals, ensuring timely completion of modules.
- **Mentorship:** Seek feedback from senior engineers at Xenon Stack and improve code quality.